WEEK 8 · LECTURE L1 · UNIT III

Dashboard Design: Perception, Attention and Restraint

What the eye sees first, what the mind can hold, what an executive should see in ten seconds, and what to leave out.

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DSEH001 · Digital Transformation in Human Resource Systems · M.Com. PGCF 2025 · Exam preparation edition · Updated 25 October 2026

AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Apply cognitive load, working memory, preattentive processing and Gestalt principles to dashboard design.
2. Choose chart types with Cleveland and McGill, and apply Tufte’s data ink and Few’s definition of a dashboard.
3. Distinguish strategic, operational and analytical dashboards, and set thresholds with signal detection in mind.
4. Explain dashboard theater and the counter metric discipline, and redesign a dashboard around decisions.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcome: CLO 4.

COMMIT BEFORE THEORY

Commit to an answer first

A CHRO’s dashboard has 14 tiles, each with a different chart type and color. She says she never uses it. What is the most likely cause?

A

She lacks data literacy

B

It overloads working memory: nothing tells her where to look or what to decide

C

The data are out of date

D

Dashboards are not useful for leaders

EXAM LENS

The answer is B.

SWELLER (1988); MILLER (1956); COWAN (2001)

Cognitive load: the mind’s narrow channel

Three kinds of load

Intrinsic (the content), extraneous (poor design: decoding legends, matching colors across tiles, hunting for the point), germane (real understanding). Remove the extraneous.

Working memory holds very little

Miller suggested about seven items; Cowan’s later review put the limit nearer four chunks. Beyond a few, viewers stop comparing and start skimming.

Design rules

Few key metrics per view; group related items; visual hierarchy; progressive disclosure: summary first, detail on demand.

EXAM LENS

Every element that does not help the decision competes with it. Five tiles is already near the limit; fourteen guarantee nothing is compared. Cite Cowan as the update to Miller.

WHAT THE EYE SEES BEFORE IT THINKS

Preattentive features and Gestalt grouping

Preattentive features

Color, size, shape, orientation, movement and position are noticed in a fraction of a second, before focused attention (Healey, Booth and Enns, 1996).

Gestalt grouping

Proximity, similarity and enclosure make items read as belonging together; figure and ground make the important item stand out.

Design rule

Use one preattentive cue to mark what matters; if everything is highlighted, nothing is. Color is a signal, not decoration.

CLEVELAND AND MCGILL (1984)

Choosing the chart

Encoding, most to least accurate	Typical chart
Position on a common scale	Bar, dot and line charts
Position on unaligned scales	Small multiples
Length, direction, angle	Pie slices by angle
Area, volume, curvature	Bubbles
Shading, color saturation	Heat maps

EXAM LENS

For comparing values use position (bars, dots); avoid angle, area and shade. Show attrition by location as sorted bars, not a pie. Based on laboratory tasks; real dashboards also need speed and memorability.

RESTRAINT · TUFTE (1983); FEW (2006)

Data ink, and what a dashboard is

Data ink

Maximize the share of ink that shows data; erase non data and redundant ink. Decoration, 3D effects and heavy grids are chartjunk. Some studies show embellishment can aid memory, so minimalism is not always best.

Few’s definition

A visual display of the most important information needed to achieve objectives, on a single screen, seen at a glance. If a CHRO must scroll, the dashboard has already failed.

The ten second test

Can the viewer say what is good, what is bad and what to do in ten seconds? Hierarchy (most important largest and first), restraint, the right chart.

EXAM LENS

A dashboard is for monitoring and deciding, not for showing everything known. The teaching note’s 5:3:1 rule says the same: five seconds to grasp the purpose, three clicks to detail, one clear next action.

MATCH THE DASHBOARD TO ITS USER

Three types, and thresholds

Type	For whom	What it shows
Strategic	Executives	A few KPIs, monthly or quarterly
Operational	Managers	Daily work, updated often
Analytical	Analysts	Data to explore and filter

EXAM LENS

Do not mix types on one screen. Signal detection: too sensitive and everything is red (alert fatigue); too insensitive and real problems are missed. The teaching note’s guide: green over 75% of the time, amber 10 to 20%, red under 5%. Use descriptive labels that give information scent (Pirolli and Card, 1999).

DASHBOARD THEATER

Pair every target with a counter metric

Target	Counter metric that reveals gaming
Chatbot deflection	Resolution rate
Time to fill	Quality of hire
Course completion	Assessment scores and on the job behavior
Low attrition	Regretted exits among high performers

EXAM LENS

Signs of theater: many tiles with no decision attached, live updates nobody acts on, only numbers that look good. Defenses: a counter measure beside each headline, trend and target not a lone number, an owner and a decision for every tile (Campbell, 1979).

REDESIGN

Keep three, cut the rest, with reasons

Keep

One lagging outcome (attrition rate with a reference line); one leading driver (share of staff under weakly rated managers); one equity or compliance tile (the adverse impact ratio of any algorithm in use).

Cut

Cumulative totals (vanity); metrics with no owner (theater); duplicates; activity counts with no outcome link (training hours, logins); individual rankings (surveillance creep).

Course framing.

APPLY

A fourteen tile dashboard for the Board

Teaching case (Course framing)

A Mumbai retail bank’s HR dashboard has 14 tiles: four pie charts, a 3D bar chart, a heat map of engagement by branch in eight shades, a speedometer for attrition and a running count of training hours. Board members say they cannot tell whether HR is doing well.

Work it through

1. Which tiles fail Cleveland and McGill’s ranking, and why?
2. What does cognitive load theory say about 14 tiles?
3. Which tile is a vanity number?
4. Redesign: which three tiles survive, and what decision does each support?

A FRAMEWORK FOR ANY CASE

For any real dashboard ask: what decision was it built for, and who owns it? How many tiles, which leading and which lagging? Is there a counter metric? Who can read and question it? What changed because it exists?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
More tiles give a fuller picture	More tiles add extraneous load; hold what matters for a decision
Pie charts are good for comparison	Angle and area are judged less accurately than position
Color makes a dashboard engaging	Color is a preattentive signal; spend it on one thing
A red light everywhere means vigilance	It means alert fatigue; recalibrate thresholds

COMPARE AND EVALUATE

Six lenses on dashboard design

Lens	Use it to	Watch out for
Sweller; Cowan	Limit load and tile count	Chunking raises capacity
Healey et al.	Direct attention	Overuse cancels the signal
Cleveland and McGill	Choose encodings	Laboratory tasks
Tufte	Remove chartjunk	Embellishment can aid memory
Few	Define and test a dashboard	Interactive drilldowns blur the single screen rule
Campbell	Anticipate gaming	Counter metrics can be gamed too *

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

Critique an HR dashboard with 12 tiles, including pie charts, a speedometer and a training hours counter. Using cognitive load theory, Cleveland and McGill's findings and Few's definition of a dashboard, propose a redesign and identify one way it could still mislead. **20 marks**

A strong answer: applies cognitive load and preattentive processing; uses Cleveland and McGill to choose charts; applies Few's single screen standard and ties tiles to decisions; names a residual risk such as Campbell's law.

"Dashboard theater": explain the concept and how counter metrics address it. **20 marks**

A strong answer: links to Goodhart and Campbell; gives target and counter metric pairs; evaluates residual risks.

Why should a CHRO dashboard contain few tiles? **10 marks**

A strong answer: uses working memory research (Miller; Cowan) and Few's definition.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Pass the ten second test.**
   Hierarchy, restraint, the right chart.

2. **Position beats angle, area and shade.**
   Bars and dots for comparison.

3. **Fewer tiles, each tied to a decision and an owner.**
   Working memory holds about four chunks.

4. **Pair every target with a counter metric.**
   Theater is the dashboard nobody acts on.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Kavanagh, M. J., & Johnson, R. D. (2018), Chapter 14 (HR metrics, workforce analytics, reporting and dashboards).
- Angrave, D., et al. (2016). *HRMJ, 26*(1).
- An Indian HR dashboard or data culture case.

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Campbell, D. T. (1979). Assessing the impact of planned social change. *Evaluation and Program Planning, 2*(1), 67 to 90.
- Cleveland, W. S., & McGill, R. (1984). Graphical perception. *Journal of the American Statistical Association, 79*(387), 531 to 554. https://doi.org/10.1080/01621459.1984.10478080
- Cowan, N. (2001). The magical number 4 in short term memory. *Behavioral and Brain Sciences, 24*(1), 87 to 114. https://doi.org/10.1017/s0140525×01003922
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- Miller, G. A. (1956). The magical number seven, plus or minus two. *Psychological Review, 63*(2), 81 to 97. https://doi.org/10.1037/h0043158
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- Sweller, J. (1988). Cognitive load during problem solving. *Cognitive Science, 12*(2), 257 to 285. https://doi.org/10.1207/s15516709cog1202_4
- Tufte, E. R. (1983). *The visual display of quantitative information*. Graphics Press.